Exact recognition of common quadratic chirps

Version 2 with supplementary bit cost appendix

11 October 2026

1 Purpose and main theorem

This note gives a deterministic, exact membership test for finite-field signals that share one unknown quadratic phase and have at most a specified number of linear frequency modes. A cyclically centered determinant is a nonzero polynomial whose roots include every valid common phase. Each surviving phase is then checked against the entire raw input. On acceptance, a quadratic Gauss-row calculation produces the complete discrete Fourier transform.

The cyclic formulation permits every rank bound $1 \le r < N$; the former interior-window assumption $N \ge 3r$ is unnecessary. This is a scope refinement of the same recognition method. Prony recovery, Toeplitz predictor recurrences, short convolution, FFT pruning, and quadratic Gauss identities are classical ingredients. Historical priority of the exact combined recognition contract remains unresolved.

Theorem 1. Let $N \ge 2$, let p be a supplied prime with $N \mid (p-1)$, and let $\omega \in F_p$ be supplied with certified exact order N. Let $1 \le r < N$. Given all N coordinates of $x \in F_p^N$, define the target family by

$$x_j = \omega^{a j^2} \sum_{\ell=1}^{s} \gamma_\ell \omega^{b_\ell j}, 0 \le j < N, 0 \le s \le r. \qquad (1)$$

Here $a, b_\ell \in \mathbb{Z}/N\mathbb{Z}$, the listed b_ℓ are distinct, and the listed $\gamma_\ell \in F_p$ are nonzero. The empty sum represents zero. There is a deterministic exact algorithm that accepts precisely this family, returns one fully verified representation on acceptance, and then returns all positive-sign unnormalized DFT values

$$\hat{x}_k = \sum_{j=0}^{N-1} x_j \omega^{k j}, 0 \le k < N. \qquad (2)$$

Rejection certifies nonmembership in (1); it does not supply the rejected vector's transform. Put $\kappa = \log_2 3$. For arbitrary admissible lengths the algorithm uses

$$O\left(N r^{2+\kappa} + r^{2+3\kappa}\right) \qquad (3)$$

field additions and multiplications. When N is dyadic, a separate refinement uses

$$O\left(N r^3 \log(r+2) + r^{2+3\kappa}\right). \qquad (4)$$

Both routes use $O(1+r^4)$ scalar field inversions and $O(N+r^4)$ live field elements. All raw reads, coefficient processing, supplied-root tables, failed candidates, verification work, and accepted output writes are charged. Integer index arithmetic, gcd and modular inverses of integers are accounted for separately from these field-operation counts. Field/root discovery, a bit-complexity bound, numerical robustness, optimized implementation timings, and constant-round parallelism are outside the theorem.

2 Circular centering and the phase polynomial

Define $x_j^\circ = x_{j \bmod N}$ for every integer j, including negative indices. This is a lookup into the supplied data, not an extra observation. A model term is itself N-periodic: replacing j by $j+N$ changes its exponent by $2aNj + aN^2 + b_\ell N$, a multiple of N. Thus a valid model matching the raw input matches this periodic continuation.

Read all coordinates first. Accept the zero vector with its empty representation and zero transform. Otherwise choose any center c with $x_c \ne 0$. Set

$$f(d) = \frac{d(d-1)}{2}, A_c(t)_{ij} = x_{c+i-j}^\circ t^{f(|i-j|)}, 0 \le i, j \le r, \qquad (5)$$

and let $E_c(t) = \det A_c(t)$. The fraction in f is an integer formula, so no field division by two is involved.

Lemma 2. The polynomial E_c is nonzero, has degree at most $D = \binom{r+2}{3}$, and vanishes at $t = \omega^{-2a}$ for every phase a in a representation (1). Every leading k-square minor of A_c has constant term x_c^k.

Proof. For integer d, $f(d) \geq 0$, and $f(d) = 0$ precisely for $d = 0, 1$. Consequently $A_c(0)$ is lower bidiagonal, with diagonal x_c and subdiagonal x_{c+1}°. It follows that

$$E_c(0) = x_c^{r+1} \neq 0. \qquad (6)$$

The same argument applies to every leading square block. Repetition of coordinates under cyclic lookup has no effect on this exponent pattern or on the unique degree-zero determinant contribution.

For a determinant term indexed by a permutation σ of $\{0, \ldots, r\}$, its exponent is

$$\sum_{i=0}^{r} f(i - \sigma(i)) = \sum_{i=0}^{r} i^2 - \sum_{i=0}^{r} i\,\sigma(i). \qquad (7)$$

Strict rearrangement maximizes (7) at reversal $\sigma(i) = r - i$, giving

$$D = 2\sum_{i=0}^{r} i^2 - r\sum_{i=0}^{r} i = \frac{r(r+1)(r+2)}{6}. \qquad (8)$$

Every minor also has degree at most D: extend a partial row-column matching to a full permutation, adding only nonnegative exponents. No characteristic or distinct-coordinate assumption enters this bound.

For a genuine phase a, let $y_j = x_j^{\circ} \omega^{-a j^2}$ on all integers and put $h = c - r$. It is an s-term exponential sequence. At $t_a = \omega^{-2a}$,

$$x_{h+i+j}^{\circ} t_a^{\binom{i+j}{2}} = y_{h+i+j}\, \omega^{a h^2 + a(2h+1)(i+j)}. \qquad (9)$$

The extra factor separates into nonzero row and column multipliers, so the $(r+1)$-square matrix in (9) has rank at most $s \leq r$. Column reversal and Laurent row-column scaling give

$$E_c(t) = (-1)^{r(r+1)/2} t^{-E_{min}} \det\left(x_{h+i+j}^{\circ} t^{\binom{i+j}{2}} \right)_{i,j=0}^{r}, \qquad (10)$$

where $E_{min} = (r+1)r(r-1)/2$. Indeed, after column reversal use $\binom{r+i-j}{2} = \binom{r}{2} + r(i-j) + f(i-j)$; the row and column powers total E_{min}. Equation (10) is a Laurent identity, evaluated only at $t_a \neq 0$. It proves $E_c(t_a) = 0$, even when $h < 0$ or indices wrap. This completes the proof.

The degree bound is attained for this particular polynomial. If every $x_j = 1$, a Vandermonde factorization yields

$$E_c(t) = \prod_{k=1}^{r} (1 - t^k)^{r+1-k}. \qquad (11)$$

To see this, factor $t^{f(i)}$ from row i and $t^{f(-j)}$ from column j; the remaining entries are $(t^{-j})^i$. The powers extracted from its Vandermonde factors cancel the initial powers. The degree is $\sum_{k=1}^{r} k(r+1-k) = D$, with leading coefficient ± 1 in every characteristic. This proves degree sharpness for E_c, not optimality over other certificates or a maximal number of distinct phase roots.

3 The phase quotient and exact recovery

Put

$$L=\frac{N}{gcd(2,N)},Q=min(D,L). \qquad (12)$$

The points ω^{-2a}, $0\le a<L$, are distinct. If N is even, the change $a\mapsto a+N/2$ can be absorbed by $b_\ell\mapsto b_\ell+N/2$, because j^2-j is even. Hence this canonical range covers every representation. It removes only a universal redundancy; more signal-dependent ambiguities may remain. For example, with $N=3$, $p=7$, $\omega=2$, $r=2$, the vector $(0,1,3)$ belongs at all three canonical phases.

Since E_c is nonzero, at most Q phases survive its root test. The usual polynomial root bound applies even if $D\ge p$. It is possible for every phase point to be a root. Reducing E_c modulo t^L-1 can accelerate evaluation, but the reduced polynomial may be zero and must not replace the original nonzero certificate.

3.1 The all length candidate test

For each surviving phase a, dechirp all raw samples by $y_j=x_j\omega^{-aj^2}$. Read the first $2r$ entries of the periodic continuation y°; since $r<N$, this uses fewer than $2N$ lookups. The following test is deterministic.

Step 1. Compute $s=rank\,H_r$, where $H_r=\left(y^\circ_{i+j}\right)_{0\le i,j<r}$. If $s=0$, require the entire raw vector to be zero. Otherwise continue.

Step 2. Require the leading s-square Hankel block to be nonsingular. Solve for the monic node polynomial $P(z)=\sum_{j=0}^{s}P_jz^j$, with $P_s=1$, from $\sum_{j=0}^{s}P_j\,y^\circ_{i+j}=0$ for $0\le i<s$.

Step 3. Evaluate P at every supplied grid point ω^b, $0\le b<N$. Require exactly s distinct grid roots $\lambda_\ell=\omega^{b_\ell}$.

Step 4. Require every ordinary unwrapped residual to vanish:

$$R_k=\sum_{j=0}^{s}P_j\,y_{k+j}=0,0\le k<N-s. \qquad (13)$$

Step 5. Solve the first s moments $y_j=\sum_\ell\gamma_\ell\lambda_\ell^j$, $0\le j<s$, using the nonsingular Vandermonde matrix. Omit zero weights and return the resulting representation.

Soundness. The constructed exponential sum agrees with the first s entries and satisfies the same monic recurrence. Equation (13) propagates equality to all N raw entries by induction. Thus a misleading periodic prefix cannot cause false acceptance. Even when $s=N-1$, the one residual checks the final raw coordinate after $N-1$ initial moments. A zero prefix alone never certifies a nonzero raw input.

Completeness. For a true s-mode sequence, $H_r=V_r\,diag(\gamma)\,V_r^T$, where V_r has full column rank s, its transpose has full row rank s, and all weights are nonzero. Hence $rank\,H_r=s$. Its leading block has determinant $\left(det\,V_s\right)^2\prod_\ell\gamma_\ell\neq0$. The solved polynomial is $\prod_\ell(z-\lambda_\ell)$, so the grid and residual tests pass and the moments recover the weights. This argument remains valid when $2r>N$, because a genuine model has exactly the periodic continuation being used.

Dense prefix rank, recurrence and Vandermonde solves cost $O(r^3)$ additions/multiplications and $O(r)$ inversions per candidate, including failed candidates. Full residual checking, rather than an extra uncharged $O(Nr)$ synthesis, will supply the all-length fast verification bound.

3.2 The dyadic recovery alternative

For dyadic N, ordinary Berlekamp-Massey on the $2r$ periodic-prefix values uses $O(r^2)$ work and $O(r)$ scalar inversions. A failed finite prefix can have order as large as $2r$; that still fits these costs. Reject if the

output order s exceeds r. Require a connection polynomial $C(z)=1+C_1 z+\cdots+C_s z^s$ of degree exactly s and exactly s distinct grid roots. For a true sequence,

$$C(z)=\prod_{\ell=1}^{s}\left(1-\lambda_\ell z\right). \qquad (14)$$

A root ω^v corresponds to node $\lambda=\omega^{-v}$, rather than ω^v. This convention differs from the monic node polynomial in Section 3.1. Compute

$$B(z)=C(z)\left(\sum_{j=0}^{s-1} y_j z^j\right) \bmod z^s, \gamma_\ell=-\frac{\lambda_\ell B\left(\lambda_\ell^{-1}\right)}{C'\left(\lambda_\ell^{-1}\right)}. \qquad (15)$$

The generating function is $\sum_\ell \gamma_\ell/\left(1-\lambda_\ell z\right)$. Multiplying it by C and evaluating at a reciprocal node proves (15); differentiation contributes the factor $-\lambda_\ell$. Distinct roots make the denominators nonzero. Quadratic polynomial arithmetic and Horner evaluations cost $O\left(r^2\right)$ work and $O(r)$ inversions. Finally synthesize and compare all N raw dechirped entries exactly. The zero-order case also requires zero full synthesis. Completeness follows from the invertible leading Hankel block and $2r\geq 2s$; soundness follows from full synthesis.

4 Computing the eliminant with unit Toeplitz pivots

Let R be the ring of polynomials over F_p modulo t^{D+1}. All coefficient arithmetic can be performed in R, since the final determinant has degree at most D. Write $a_d=x_{c+d}^{\circ} t^{f(d)}$ and $T_k=\left(a_{i-j}\right)_{i,j=1}^{k}$, with $\det T_0=1$. Lemma 2 makes every $\det T_k$ a unit in R.

For completeness, the inherited nonsymmetric predictor recursion is given explicitly. Let $u^{(k)}, v^{(k)}$ be indexed from 0 through k, with $u_0=v_0=1$, satisfying

$$\sum_{j=0}^{k} a_{i-j}u_j=0, \sum_{j=0}^{k} a_{j-i}v_j=0, 1\leq i\leq k. \qquad (16)$$

Their common error term is

$$e_k=\sum_{j=0}^{k} a_{-j}u_j=\sum_{j=0}^{k} a_j v_j=\frac{\det T_{k+1}}{\det T_k}, e_k(0)=x_c. \qquad (17)$$

The equalities follow from Schur complements and transposition. Initialize $u=v=(1)$, $e=a_0$, and $\Delta=a_0$. At step $k=1,\ldots,r$, use the previous length-k vectors to form

$$A_k=a_k+\sum_{j=1}^{k-1} a_{k-j}u_j, B_k=a_{-k}+\sum_{j=1}^{k-1} a_{j-k}v_j, \qquad (18)$$

$$\alpha_k=-A_k/e, \beta_k=-B_k/e. \qquad (19)$$

Simultaneously update $u'_j=u_j+\alpha_k v_{k-j}$ and $v'_j=v_j+\beta_k u_{k-j}$ for $1\leq j<k$, setting $u'_0=v'_0=1$, $u'_k=\alpha_k$, and $v'_k=\beta_k$. Then set

$$e'=e+\alpha_k B_k=e-A_k B_k/e, \Delta'=\Delta e'. \qquad (20)$$

The reversed old predictor contributes zero to the first $k-1$ equations and e to the new final equation. Thus (19) cancels its residual and preserves (16). Expanding the new error gives (20). At completion $\Delta=E_c$ in R and therefore as a degree-at-most-D polynomial.

Every reciprocal starts from the same scalar inverse x_c^{-1}. Newton doubling $g\leftarrow g(2-eg)$ lifts a reciprocal to twice its precision, since $1-eg_{new}=(1-eg)^2$. This identity uses no division by two. Karatsuba multiplication costs $M(m)=O\left(m^\kappa\right)$; each unit-series reciprocal costs $O(M(D+1))$. The $O\left(r^2\right)$ ring operations and r reciprocal calculations therefore cost

$$O\left(r^2(D+1)^{\kappa}\right) \qquad (21)$$

field additions/multiplications, one scalar inverse, and $O(r(D+1))$ live coefficients. The construction has r predictor stages and $O(\log(D+1))$ precision doublings for each reciprocal. A small scalar-inversion count does not imply constant sequential depth.

5 Paying for evaluation and full verification

5.1 Arbitrary admissible lengths

Let $F(z)=\sum_{j=0}^{s} F_j z^j$ and $b=s+1\le N$. Split the output indices into blocks $k=K+u$, $0\le u<b$, allowing a shorter last block. The integer identity

$$u\,j=\binom{u}{2}+\binom{j+1}{2}-\binom{u-j}{2} \qquad (22)$$

gives

$$F\left(\omega^{K+u}\right)=\omega^{\binom{u}{2}}\sum_{j=0}^{s}\left(F_j\omega^{Kj+\binom{j+1}{2}}\right)\omega^{-\binom{u-j}{2}}. \qquad (23)$$

Each block is a length-b by length-$(2b-1)$ ordinary convolution, computed by a constant number of balanced Karatsuba products in $O(b^{\kappa})$ operations. All scalings cost $O(b)$ per block; the kernel is reusable, and changing K by b updates the coefficients with precomputed ω^{bj}. The total is $O(N b^{\kappa-1})$ work and $O(N+b)$ storage. No square root of ω, extension field, or smooth transform length is needed.

For the residuals (13), convolve y with the reversed coefficients $(P_s,\dots,P_0)$. The residual R_k is convolution coefficient $k+s$. Partition y into length-b chunks, multiply each by this fixed length-b kernel, and overlap-add. The same $O(N b^{\kappa-1})$ bound pays for all $N-s$ residuals and every raw coordinate. The sole size condition is $b=s+1\le N$, which follows from $r<N$.

Horner evaluation of E_c on all L phase points costs $O(N(D+1))$. Thus each surviving candidate, including every failure, costs $O\left(N(r+1)^{\kappa-1}+r^3\right)$ work and $O(r)$ inversions.

5.2 The dyadic refinement

When N is a power of two, a radix-two DFT with at most q nonzero input coefficients can skip all-zero branches. At recursion depth d, at most $\min(2^d,q)$ branches of length $N/2^d$ are active. Its combine work is bounded by

$$O\left(N\sum_{d=0}^{\log_2 N-1}\min\left(1,q/2^d\right)\right)=O(N\log(q+2)). \qquad (24)$$

This uses only roots of orders dividing N. Applying it to the degree-s connection polynomial costs $O(N\log(r+2))$. Applying it to the at-most-s frequency weights produces full synthesis at the same cost. Together with Section 3.2, this gives $O\left(N\log(r+2)+r^2\right)$ per surviving phase.

For phase enumeration, reduce E_c modulo t^L-1 in $O(D+L)$ work, paying for each coefficient. The reduced array has support at most $\min(D+1,L)$. A length-L pruned DFT with root ω^{-2} scans the phase points; $L=1$ is evaluated directly. The cost is $O\left(D+N\log\left(\min(D+1,L)+2\right)\right)$. If reduction yields the zero polynomial, all phase points survive and are still charged. These dyadic bounds are not asserted for arbitrary N.

6 All accepted Fourier outputs from one Gauss row

After accepting one representation, form the shared row

$$G_a(v)=\sum_{j=0}^{N-1}\omega^{a j^2+v j}. \qquad (25)$$

Then $\hat{x}_k=\sum_{\ell=1}^{s}\gamma_\ell G_a(k+b_\ell)$, with indices modulo N. The row costs $O(N)$ field operations and the final combination costs $O(N s+N)$, including output writes. The construction below covers nonunit phases and even composite lengths.

Use the representative $0\le a<N$, set $d=gcd(a,N)$, $m=N/d$, and $A=a/d$. For $a=0$, set $d=N$, $m=1$, $A=0$. Write $j=u+v m$. Summing the geometric progression over v shows that $G_a(c)=0$ unless $d \mid c$. If $c=d w$, then

$$G_a(d w)=d\,T(w), T(w)=\sum_{u=0}^{m-1}\eta^{A u^2+w u}, \eta=\omega^d. \qquad (26)$$

For $m=1$, this is the N-spike row: $G_0(0)=N$ and the other entries vanish. Otherwise A is invertible modulo m.

If m is odd, a shift completing the square gives

$$T(w)=T(0)\eta^{-w^2(4A)^{-1}}, \qquad (27)$$

where the inverse is modulo the integer m. Compute $T(0)$ by one explicit length-m sum; a table of powers of η then gives the row.

If $m=2q$ is even, A is odd. Translation by q multiplies the summand by $(-1)^{Aq+w}$, so $T(w)=0$ unless w has the parity of q. Set $w_0=q \bmod 2$ and compute $T(w_0)$ directly in $O(m)$ field work. For an allowed w, solve

$$u=\frac{w-w_0}{2}A^{-1}(\bmod q), T(w)=T(w_0)\eta^{-Au^2-w_0 u}. \qquad (28)$$

When $q=1$, use $u=0$ without an inverse modulo one. The shift $j\mapsto j+u$ proves (28). Changing u by q changes the exponent by a multiple of $2q$, because $Aq+w_0$ is even. Thus the chosen lift does not matter. Since $p>N\ge 2$, characteristic two is excluded from the recognition theorem, and the parity cancellation is valid.

Integer gcd and modular-inverse calculations in (26)–(28) are separate from scalar inversion in F_p. This direct base-row construction does not assume a closed-form square root of a Gauss sum or an uncharged Gauss-sum oracle.

7 The complete cost ledger

Process candidates one at a time and stop at the first fully verified representation. Worst-case bounds still pay for all Q surviving phases. With $D=\binom{r+2}{3}$, $Q=min(D,L)$, the all-length work is

$$O\left(r^2(D+1)^{\kappa}+N(D+1)\right) \qquad (29)$$

plus

$$O\left(Q\left(N(r+1)^{\kappa-1}+r^3\right)+N r+N\right). \qquad (30)$$

The dyadic work is

$$O\left(r^2(D+1)^{\kappa}+D+N\log\left(min(D+1,L)+2\right)\right) \qquad (31)$$

plus

$$O\left(Q\left(N\log(r+2)+r^2\right)+N r+N\right). \qquad (32)$$

The terms have the following meanings.

Raw input and supplied roots. All N reads, the zero test, nonzero center, root-power tables, and $O(N)$ preprocessing.

Polynomial certificate. Coefficient creation, the unit Toeplitz recursion and coefficient writes in (21).

Phase scan. Horner at arbitrary length; paid reduction and support-pruned scan at dyadic length.

Every surviving phase. All dechirping, prefix recovery, root checks, weights, and full-input residual or synthesis checks, including failures.

Accepted transform. One complete Gauss row and $O(N r+N)$ combination and output work.

Because $D=O(r^3)$ and $Q\le D$, (29)–(30) imply (3): the r^6 dense-prefix contribution is dominated by $r^{2+3\kappa}$, and the scan term $N r^3$ by $N r^{2+\kappa}$. Equations (31)–(32) imply (4), with the r^5 recovery contribution also dominated by $r^{2+3\kappa}$.

The Toeplitz construction uses one scalar inverse. A candidate needs $O(r)$ further scalar inversions, hence the total is $O(1+Q r)=O(1+r^4)$. Live storage is

$$O\left(N+r(D+1)+r^2\right)=O\left(N+r^4\right). \qquad (33)$$

This accounts for the raw and output arrays, coefficient/predictor storage and one candidate's matrices or recurrence. Scalar inversion, field arithmetic, coefficient storage and sequential stages are distinct quantities. Dense elimination may have $O(r)$ pivot stages; dyadic transforms have logarithmic recursion depth.

For fixed r, the all-length bound is $O_r(N)$. More generally, its sufficient growing-rank regime for $o(N\log N)$ field additions/multiplications is $r=o\left((\log N)^{1/(2+\kappa)}\right)$. This is a statement about membership recognition and accepted structured outputs. It neither computes transforms of rejected inputs nor improves a general-purpose FFT theorem. For r comparable to N, the conservative bounds are expensive.

8 Classical antecedents and the unresolved comparison

The conceptual division matters for evaluating the contribution. The certificate in Lemma 2 has a data-independent nonvanishing guarantee once any nonzero center is selected; it is joined to a supplied-grid, full-input acceptance and rejection contract. The reconstruction and arithmetic tools underneath it have established antecedents. The cyclic extension removes a witness-placement restriction; it neither changes the asymptotic exponents nor creates a separate discovery claim.

Prony and dechirping. Once a phase is fixed, recurrence recovery, a node polynomial, and a Vandermonde amplitude solve are ordinary exponential-sum reconstruction. Plonka, Stampfer and Keller [2], Section 4 and Theorem 4.1, treat a known nonvanishing factor and a supplied quadratic coefficient on a uniform grid. Their Theorem 5.3 treats varying unknown quadratic coefficients on a square-root sampling grid; Section

6.2 uses a fixed coefficient with cosine and sine data. These are distinct input interfaces. Cuyt and Lee [3], Section 5, equation (31) and Theorem 4, also supply a common Gaussian width before exponential reconstruction. Their cited Peter thesis pages 85–91 remain outside the inspected evidence.

Shared-parameter elimination. Agostini, Améndola and Ranestad [1], Section 4.3, equations (79)–(81), use trial variance to transform moments, obtain a Hankel determinant polynomial, and then apply Prony/Vandermonde recovery. This establishes the shared-parameter elimination framework. The present ordinates are fixed raw finite-field samples multiplied by binomial monomials; moment-domain statements alone do not settle the exact same certificate and decision contract. The attribution to Lindsay's Theorem 5B in [1] does not replace inspection of Lindsay's original hypotheses.

Toeplitz and convolution. The paired predictors and nonsingular-leading-minor condition are classical nonsymmetric Levinson/Schur machinery, explicitly developed in Gutknecht and Hochbruck [4], Section 2, equations (2.7)–(2.10) and Algorithm 1. Here the cyclic normalization supplies the unit-pivot guarantee in a truncated polynomial ring and the ledger pays for its arithmetic. The blocked chirp-z and overlap-add mechanisms in Section 5 are likewise classical, with the sectioned chirp-z precedent recorded in Rabiner, Schafer and Rader [8]. This note claims neither a new general Toeplitz kernel nor a new convolution or FFT identity.

Quadratic Gauss sums. Equations (26)–(28) are elementary reductions by a geometric sum and changes of variable. Including their nonunit and even-length cases is essential to the all-length contract; they are not offered as a new theory of Gauss sums.

Three original-source gaps remain: the complete body of Derin (1987) [5], the original statement and context of Lindsay (1989), Theorem 5B [6], and the complete journal body of Applebaum, Howard, Searle and Calderbank (2009) [7]. Existing abstract, indexed-passage and later-article comparisons cannot be promoted to a full primary-body read. In particular, a sparse-representation uniqueness theorem and a peak-based decoder must not be conflated with deterministic full-input recognition. Failed access is not evidence that an antecedent is absent. No first-ever or submission-ready priority claim follows from the mathematical review.

9 Review evidence and remaining limits

The cyclic-window theorem was independently reviewed on 11 October 2026 with no mathematical repair required. Its accepted scope includes $1 \le r < N$, arbitrary nonzero cyclic centers, the periodic recovery prefix, all candidate-failure costs and the separate dyadic refinement. The exact source and review bindings are recorded in the accompanying provenance file.

The independent reference controls recorded 4,250 vector/rank decisions and 15,687 per-phase decisions with no disagreement against a separately evaluated Fourier-support oracle. They included exhaustive spaces F_3^2, F_7^3 and F_5^4 for every $1 \le r < N$, followed by 1,680 random and structured cases through $N = 16$. There were 21,658 true-phase/nonzero-center checks, 436 false-phase eliminant roots rejected by full tests, 480 wrapped Laurent-identity checks, and 24 constant-window degree identities. Further controls covered 780 periodic-prefix Berlekamp-Massey/partial-fraction decisions, 67 pruned-FFT vectors, and 115 Gauss rows with 1,809 entries. Two late-spike controls rejected a zero recovery prefix followed by a nonzero raw entry.

Those are finite reference checks, not an optimized implementation of the combined polynomial Toeplitz, blocked verification, pruned-transform and Gauss pipeline. The proofs establish general correctness and the operation bounds; the counts do not establish asymptotic performance, external priority or benchmark superiority. Extra characteristic-two determinant controls concern polynomial algebra only and are not admissible recognition instances for $N \ge 2$.

All N raw entries remain necessary inputs to this stated algorithm. It supplies no sample-reduction theorem, no noisy or floating-point recognition guarantee, no unique common-phase theorem and no transform upon rejection. For $r \ge N$, every vector already has an ordinary dense Fourier representation with $a = 0$; this separate trivial branch is not a new structured fast-transform result. The case $N = 1$ is immediate. Further publication work consists of resolving the named source gaps and, if implementation performance is to be claimed, integrating and independently testing the complete optimized pipeline.

References and source access

The comparison below follows the recorded primary-source inspection and its access limits. It does not claim a new literature search. The proofs in Sections 2–7 are supplied in the note.

1. Agostini, Améndola and Ranestad, primary article, Section 4.3, equations (79)–(81). Relevant complete HTML body inspected in the source comparison. https://link.springer.com/article/10.1007/s10208-020-09469-6
2. Plonka, Stampfer and Keller, author manuscript dated 19 July 2018. Sections 4, 5.2 and 6.2; Theorems 4.1, 5.3 and 6.2. Searchable 28-page primary PDF inspected at the cited sections; rendered-page inspection was not available. https://num.math.uni-goettingen.de/~gplonka/pdfs/general-Prony-revised-sw.pdf
3. Cuyt and Lee, primary article, Section 5, equation (31) and Theorem 4. Relevant complete HTML body inspected. https://link.springer.com/article/10.1007/s11075-023-01564-3
4. Gutknecht and Hochbruck, Look-ahead Levinson and Schur algorithms for non-Hermitian Toeplitz systems (1995). Section 2, electronic pages 184–186, equations (2.7)–(2.10), Algorithm 1, and equations (2.17)–(2.20). Searchable primary author PDF inspected. https://people.math.ethz.ch/~mhg/pub/mhg-published/GutH95-NM70-laLSnHTs.pdf
5. Derin (1987). Original body remains unavailable in the source record; abstract and bibliographic evidence only. https://doi.org/10.1109/TPAMI.1987.4767880
6. Lindsay (1989). Original Theorem 5B and its surrounding hypotheses remain unread; attribution checked in [1]. https://doi.org/10.1214/aos/1176347138
7. Applebaum, Howard, Searle and Calderbank (2009). Indexed primary passages were compared previously; the complete journal body remains uninspected. https://doi.org/10.1016/j.acha.2008.08.002
8. Rabiner, Schafer and Rader (1969), The chirp z-transform algorithm. Classical sectioned chirp-z and overlap-add/save antecedent as documented in the inherited blocked-verification dependency. No fresh full-body inspection is claimed here.
9. CHIRP cyclic window and primary comparison, 11 October 2026. Author theorem, historical comparison and dependency proofs, preserved as the source package.
10. CHIRP cyclic window independent review, 11 October 2026. Scoped mathematical PASS, reviewed statement, independent reference controls and input verification, preserved as the review package.

Appendix A Supplementary bit cost contract

This appendix adds an explicit binary representation and conservative bit-cost corollary to the field-operation theorem. Theorem 1 and equations (1)–(33) are unchanged. Its exclusion of bit complexity applies to that theorem; the separate statement below uses additional representation and machine assumptions. The appendix follows the independently reviewed bit-cost statement, including its counter-width correction and block-input clarification. It does not add a novelty or performance claim.

A1 Binary input and paid access model

Let $N \ge 2$, $1 \le r < N$, and let $p > N$ be a supplied prime with $N \mid (p-1)$. Let ω be supplied as a residue, and put $B = bitlen(p) = \lfloor \log_2 p \rfloor + 1$. On this admissible prime domain p is odd and $B = \lceil \log_2 p \rceil$. No polynomial bound on p in terms of N is assumed unless explicitly stated.

Encode p, N, r as Elias gamma codes of $v+1$, in that order. For a positive integer u with binary length ℓ, this code consists of $\ell - 1$ zero bits followed by the ℓ-bit binary expansion of u. Follow the three headers by $\omega, x_0, \ldots, x_{N-1}$, each as a canonical B-bit unsigned residue in $\{0, \ldots, p-1\}$. No trailing bits are allowed. Let S be the actual input-record length. For a valid complete record,

$$S = \sum_{v \in \{p, N, r\}} \left(2\,bitlen(v+1) - 1\right) + (N+1)B = \Theta(N B). \qquad (34)$$

Use a deterministic sequential bit-cost arithmetic RAM. Copying, addition and comparison of h-bit integers cost $O(h)$; schoolbook multiplication and division cost $O(h^2)$. Access to a contiguous h-bit record at an a-bit address costs $O(h + a)$. Input is a finite record with end-of-input detection. All address calculations, counters and control work are charged. This is not an unqualified multitape-Turing-machine theorem or a fixed hardware-word theorem.

A paid reader parses gamma prefixes in geometrically increasing contiguous blocks. It compares each block with zero and locates the first one in the final block by successive splitting. Copy each header's binary payload once. The total copied or compared bits are linear in the inspected prefix; the sum of address and counter overheads is absorbed by this amount and the $O(B^2)$ header allowance. Reject impossible N or r from its decoded length or bounds before allocating arrays. Read field coordinates in B-bit records, and probe once after the final expected field to detect trailing input. Thus parsing and format checks cost $O(S + B^2)$ on the valid-parameter contract. This does not assume that arbitrary one-bit absolute-address accesses have constant cost.

A2 Parameter preparation and result meanings

Validate the header constraints and residue ranges. Generate $h_0 = 1$ and $h_k = h_{k-1}\,\omega \bmod p$ through $k = N$. Require $h_N = 1$ and $h_k \ne 1$ for $0 < k < N$; retain $h_0, \ldots, h_{N-1}$. This both certifies exact order N and constructs the supplied-root table in $O(N B^2)$ bit time. A smaller order would have appeared as an earlier return to one. No factorization of N or $p-1$, discrete logarithm or root search is used. A cached table is outside this default grammar; supporting one requires its own framing, validation and cost.

Complete the selected parameter-validation policy and inspect all raw coordinates before accepting a zero vector. On every valid prime/grid instance, the unchanged recognizer accepts exactly the common-quadratic-phase family (1). Acceptance returns a fully verified representation and all positive-sign unnormalized DFT coordinates (2), encoded as canonical B-bit residues. Rejection certifies nonmembership only; it supplies no transform of the rejected vector.

Malformed records have status INVALID_INPUT. Detected invalid bounds, order or other parameter conditions have status INVALID_PARAMETERS. Neither status is structured nonmembership. No complexity expression through nonexistent decoded N, r, B is asserted for a malformed header. Supplied-prime mode assumes primality; Section A5 gives a separately paid validator for an arbitrary supplied modulus.

A3 Supplementary corollary and unchanged ledgers

Use the same $\kappa = \log_2 3$, $D = \binom{r+2}{3}$, $L = N / gcd(2, N)$ and $Q = min(D, L)$, and set $I = 1 + Q\,r$. Collect equations (29)–(30) and (31)–(32), respectively, as

$$W_{all} = r^2 (D+1)^{\kappa} + N(D+1) + Q\left[N(r+1)^{\kappa-1} + r^3\right] + N\,r + N, \qquad (35)$$

$$W_{dyadic} = r^2 (D+1)^{\kappa} + D + N \log_2(min(D+1, L) + 2) + Q\left[N \log_2(r+2) + r^2\right] + N\,r + N. \qquad (36)$$

These are the accepted arithmetic ledgers with no term removed. Use $W = W_{all}$ at arbitrary admissible lengths; the alternative $W = W_{dyadic}$ is available only for dyadic N. Both include preparation, raw reads, coefficient processing, every failed candidate, full verification and accepted output writes.

Corollary A1. In the representation and paid-address RAM of Section A1, there is an implementation of the exact recognizer with

$$O\left(S + B^2 W + I\,B^3\right) \text{bit time}, O\left(\left(N + r^4\right) B\right) \text{working bits}. \qquad (37)$$

The corollary applies to the valid prime/grid computation, including both membership acceptance and nonmembership rejection. The optional arbitrary-modulus validation charge in Section A5 is additional.

Arithmetic and inverse costs. Every scalar product is reduced modulo p, so an unreduced product has at most $2B$ bits. A field addition or multiplication, with reduction and paid access, costs at most $O(B^2)$. Dense polynomial coefficients remain field residues throughout. For a nonzero scalar, Fermat inversion by binary exponentiation to $p-2$ uses $O(B)$ modular multiplications and costs $O(B^3)$ time and $O(B)$ scalar scratch.

There is one Toeplitz seed inverse x_c^{-1}. All Newton unit-series reciprocals reuse it as their initial scalar inverse; their polynomial operations are already in W. Every attempted phase, including failed candidates, uses $O(r)$ further scalar inversions. Divisors are tested or known nonzero before use. Hence all scalar inversions cost $O(I\,B^3)$. A constant number of integer gcd and modular-inverse calls in the accepted Gauss construction costs $O(B^3)$ in total, absorbed because $I \geq 1$. The $a=0$, $m=1$ and $q=1$ branches avoid an inverse modulo one. Negative root exponents are table lookups after integer reduction, not field inversions.

A4 Integer control work and storage

Structural indices and fixed-degree polynomial exponent expressions have $O(\log N)$ bits. Binary I/O offsets, bit-array addresses and arithmetic-loop counters may require $O(\log N + \log B)$ bits, which is $O(B)$. The broader assertion that every integer counter has $O(\log N)$ bits is not valid for unrestricted p: even with fixed N, a Fermat loop can require a $\Theta(\log B)$-bit counter. The optional trial-divisor index in Section A5 can use $O(B)$ bits. Indices and addresses are never reduced modulo p.

To see addressability, write $M = N + r(D+1) + r^2 = O(N + r^4) = O(N^4)$. A flattened $O(M\,B)$-bit allocation has address width $O(\log M + \log B) = O(\log N + \log B) = O(B)$ because $N < p < 2^B$. These addresses can be constant multiples of B bits. In particular, the number of field cells can exceed p; an address is not required to fit in one field residue.

The number of ordinary high-level control steps can be $O(W)$. Dense Toeplitz and Newton coefficient scans, Karatsuba splits and combinations, candidate rank and solve work, blocked evaluation and overlap-add residual checks fit their existing ledgers. Process predictor stages and candidate phases sequentially. For dyadic pruning, split current support by index bits, using at most $min(2^d, q)$ occupied subproblems at depth d as in (24). Build occupancy by scanning the length-N grid and this same tree, without adding an unanalysed sorting or balanced-tree routine. This pays the array traffic and metadata inside the accepted pruned-transform bound.

Ordinary address and fixed-degree integer calculations cost at most $O(B^2)$ per such step. Integer degree, padding and recursion logic implement the algorithms: D is formed by integer arithmetic, precision and

padding use doubling, and dyadic recursion uses index bits and depth counters. The real logarithms and κ in the displayed bounds are analysis notation, not runtime numerical-log oracles.

Combining $O(N B^2)$ exact-order preparation, $O(B^2 W)$ coefficient and control work, $O(I B^3)$ inversions, and $O(S + B^2)$ parsing gives (37), since $W \geq N \geq 2$. Field records and $O(B)$-bit pointers require $O(M B)$ bits. The raw input, root table, live predictors, one candidate and all accepted outputs fit this storage; scalar scratch and depth-first recursion metadata fit within the same bound. This is a valid-parameter working-space claim, not a claim about arbitrary malformed records with no defined parameter tuple.

A5 Prime validation and representation specific consequences

An optional validator for an arbitrary supplied modulus first rejects $p < 2$, then performs deterministic trial division by integers $d = 2, 3, \ldots$ while $d^2 \leq p$. Each division, square comparison and increment costs at most $O(B^2)$. This adds

$$O(\sqrt{p}\, B^2) \text{ time and } O(B) \text{ scratch}. \qquad (38)$$

This conservative time is exponential in B. The validator does not discover a suitable p or ω from N. Exact-order validation alone is not a primality test: $p = 9$, $N = 2$, $\omega = 8$ passes divisibility and order checks but is not a prime-field instance. Supplied-prime mode makes no primality-verification claim; its prime promise must not be removed silently. The optional validator must run before zero-vector acceptance.

If r is fixed and, additionally, $p = N^{O(1)}$, then $B = \Theta(\log N)$ and (37) specializes to

$$O_r\left(N (\log N)^2 + (\log N)^3\right) \text{time}, O_r(N \log N) \text{ working bits}. \qquad (39)$$

This specialization uses supplied-prime mode; the optional trial-division cost remains additional if chosen. No polynomial-size modulus assumption is made in the unrestricted-p statement (37).

The explicit fixed-width raw array contains $N B$ coordinate bits, and the accepted output emits $N B$ bits. The required full scan and full output therefore entail $\Omega(N B)$ bit traffic in this contract, hence $\Omega(N \log N)$ because $p > N$. Consequently the little-$o(N \log N)$ field-operation regime in Section 7 is not a little-$o(N \log N)$ bit-time claim for this interface. This is a representation-specific consequence, not a universal lower bound for compressed input or output, promise access, or a different problem.

No new FFT theorem, novelty, beta-dependent bound, constant-round result or optimized-timing claim follows. The Derin, Lindsay Theorem 5B and complete Applebaum-body comparison gaps in Section 8 remain open. Finite preparation checks in the supplementary review are functional controls, not an optimized integrated recognizer or a proof from timing measurements.

A6 Supplementary sources and version record

Version 2 preserves the original mathematical body and adds this appendix. The author, review and consistency packets remain unchanged. The source package records exact SHA-256 and Library-version bindings. The corrected statement in [12] controls this appendix; final integration fidelity is a separate check.

11. CHIRP Bit Contract Audit and Supplement, 11 October 2026. Original author report and preparation controls, with the counter-width correction supplied by [12].
12. CHIRP Bit Contract Independent Review, 11 October 2026. Accept with corrected counter widths and paid block-input clarification. Includes the controlling reviewed statement; the field-operation theorem remains unchanged.
13. CHIRP Working Note Consistency Review, 11 October 2026. PASS for the original nine-page note, binding its Markdown, editable Word mathematics and PDF to the accepted theorem.
14. Victor Shoup, A Computational Introduction to Number Theory and Algebra, Version 2 (2008). Theorem 3.3; Theorems 4.4–4.5; Section 10.1. Primary-text arithmetic background as inspected in [12]. The paid-address model is specified here. No fresh full-book inspection is claimed. https://shoup.net/ntb/ntb-v2.pdf